Post-bop modal-tonal AI jazz improvisation

Program(me) Notes

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COLLEGE

Now’s the time

- Back in the late 1900s I was part of a jazz workshop helping people learn to play jazz.
- One of the exercises was called "slick licks" - we would sit in a circle and “trade licks” on our instruments.
- A big barrier for jazz learners can be finding people to play with.
- Q. Can I build a software system to enable people to practice their slick licks? And specifically… can a bass player use this?!
- It should listen and respond, sometimes lead, be realistic enough to be useful.
- NB This interactive “robot” is designed for practice purposes, and it's not intended to replace a human player. Human interaction was key to slick licks!

All the things I have been* thinking about

Phrase analysis
Question and answer
Leadership & role swapping
Contour steering
Chord conditioning
Harmonic modes
Scale pitch bias
Pitch range
Swing / triplet feel
Live tempo tracking

Dynamics
Articulation
Bass pitch-class tracking
Energy arc
Motivic development
Lyrical motif re-use
Voice leading
Modal leap bonus
Stochastic thinning
Phrase breathing

Singable duration bias
Rhythmic density
Rhythmic complementarity
Register contrast
Phrase length control
Beat-matching / trading bars
Ostinato
Proactive mode
Arc timing
Phrase-shape dynamics

** Good name for a song?*

Fly me to the moon

Five giant steps in the arc:

1. **Sparse**
   - Open, sustained, establish motifs slowly
2. **Building**
   - Mixed, lines begin to flow
3. **Peak**
   - Fast, active, max rhythmic intensity
4. **Recapitulation**
   - Lyrical return, recall themes with space
5. **Resolution**
   - Slowest, long tones, fading out

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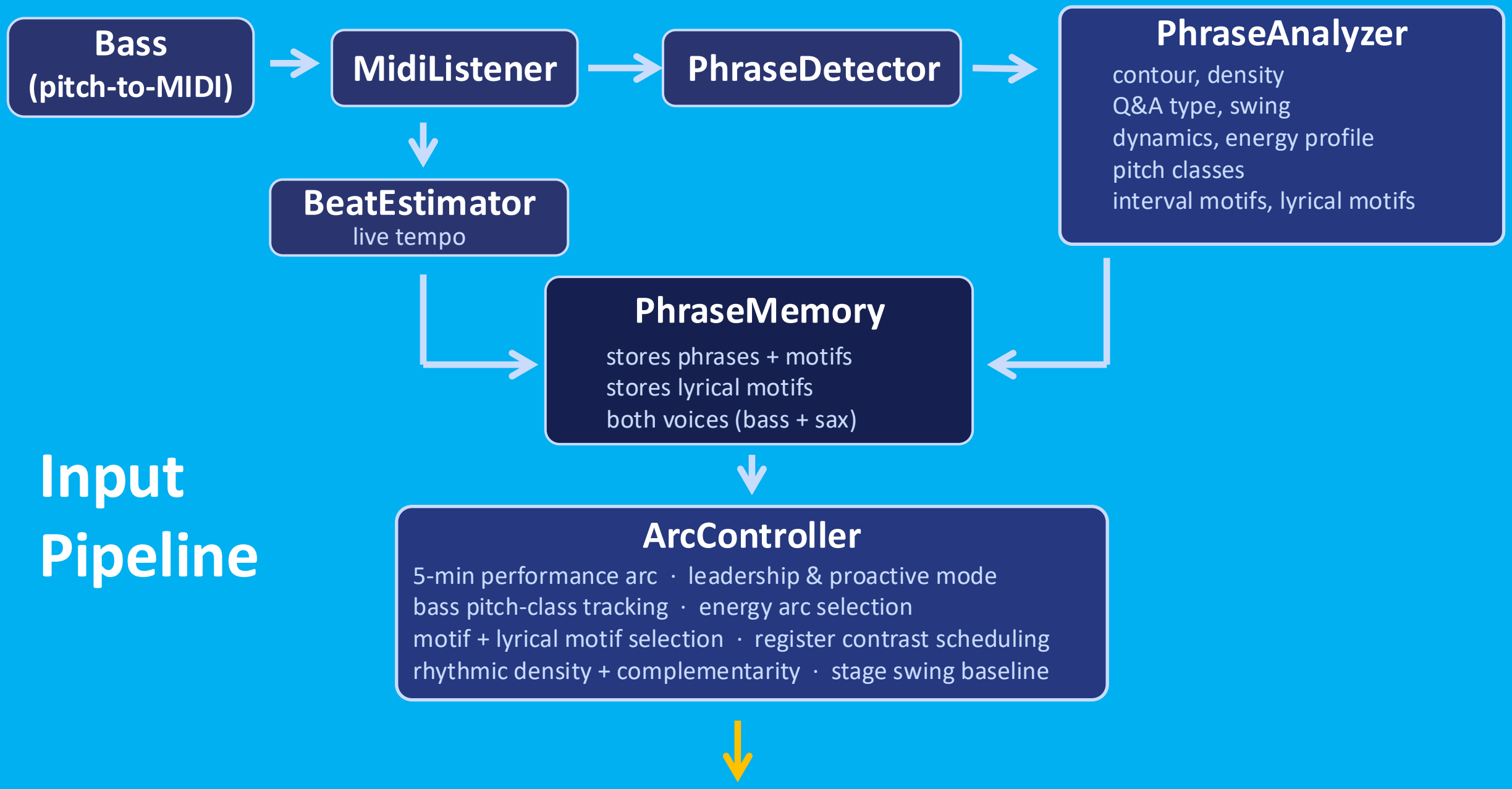
Sentimental journeys

This is inspired by:

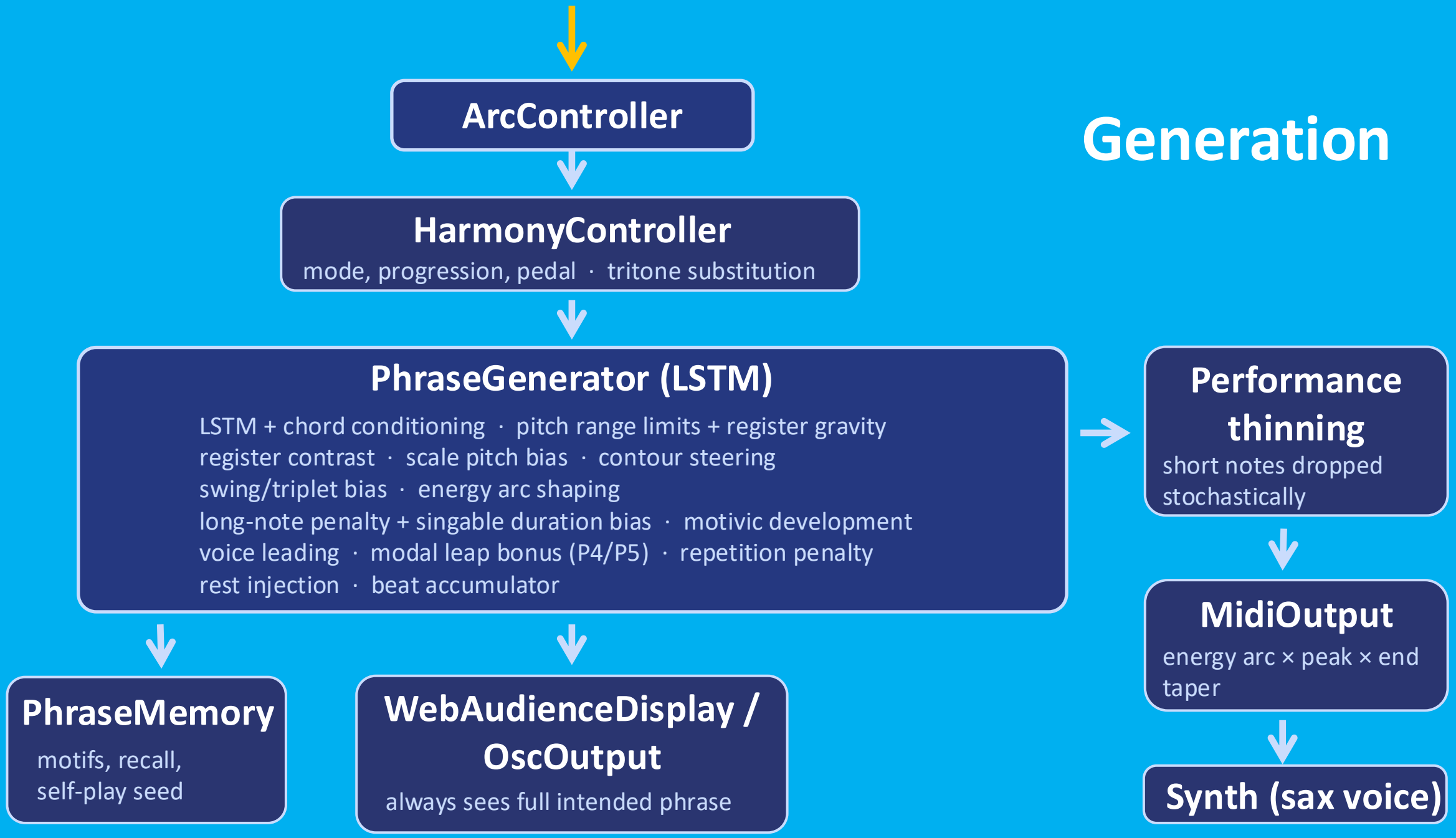
- *Post-bop modal-tonal hybrids* like:
  - **Milestones** (Miles Davis), **Little Sunflower** (Freddie Hubbard), **Maiden Voyage** (Herbie Hancock), **Footprints** (Wayne Shorter). These songs are my favourite things*.
- But also a journey through jazz:
  - Sparse opening resembles the chromatic freedom of free jazz,
  - Modal building stage reflects the Miles/Coltrane modal period,
  - Peak progression echoes hard bop,
  - Pedal resolution has the meditative quality of late Coltrane.

* John Coltrane

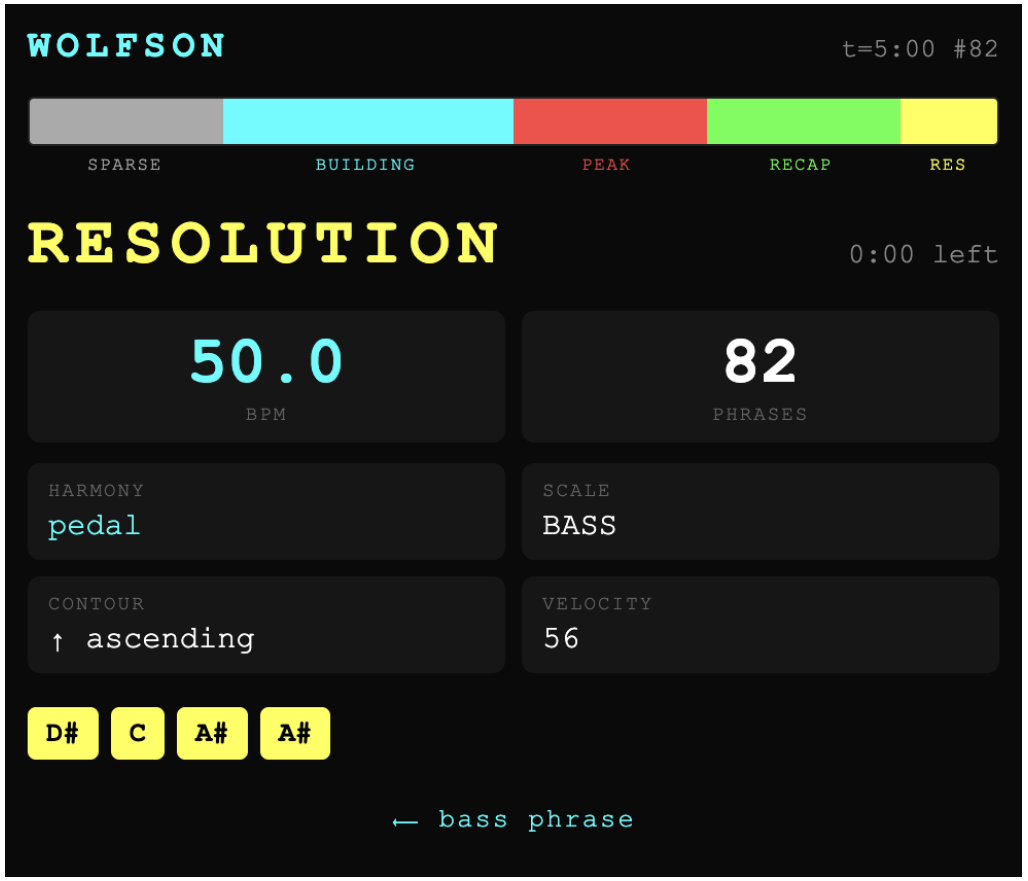
Input Pipeline

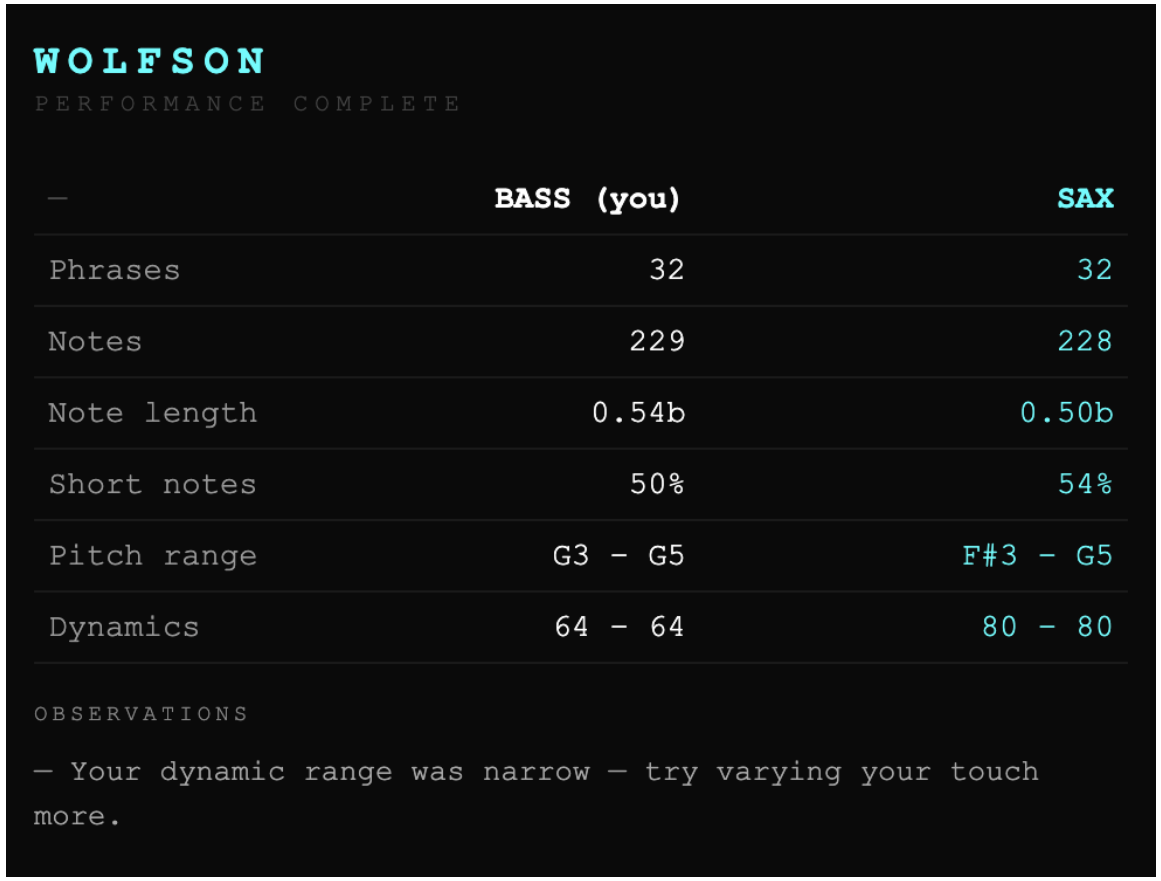


Generation


ArcController
HarmonyController
mode, progression, pedal · tritone substitution
PhraseGenerator (LSTM)
LSTM + chord conditioning · pitch range limits + register gravity
register contrast · scale pitch bias · contour steering
swing/triplet bias · energy arc shaping
long-note penalty + singable duration bias · motivic development
voice leading · modal leap bonus (P4/P5) · repetition penalty
rest injection · beat accumulator
Performance thinning
short notes dropped stochastically
MidiOutput
energy arc × peak × end taper
Synth (sax voice)
PhraseMemory
motifs, recall, self-play seed
WebAudienceDisplay / OscOutput
always sees full intended phrase

Soul eyes





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- Repo is on github.com/davidderoure/wolfson

- Acknowledgement:
  - Ray d'Inverno (for jazz, Lisp, AI, slick licks)
  - Mark d'Inverno (for jazz, AI, creativity, slick licks)
  - George E Lewis (for Voyager and Forager)
  - Robert Laidlow (for Alter-ing Ada Lovelace, and sax)
  - Hongshuo Fan (for Music Gesture Recognition)
  - Bofan Ma (for always enabling musicality)
  - Maria Kallionpää (for Climb!)
  - Wolfson Digital Research Cluster (for AIMC 2024)

- *Grand Challenges for Computing Research* – Tony Hoare

Radar of the lost archive

- The Grand Challenges Exercise of the UK Computing Research Committee (UKCRC) .
- Workshop held in Edinburgh November 2002, discussed 109 submissions.
- Archival research suggests my submission was “The creative electronic musician”. Or maybe “Interactive Electronic Musician”.
- Tony Hoare was Chair of the programme committee.

National e-Science Centre

Workshop on Grand Challenges for Computing Research

Sponsored by the UK Computing Research Committee, with support from EPSRC and NeSC

UK COMPUTING RESEARCH COMMITTEE

Panel C Submissions

5. Deep Adaption
6. Virtual Realms
7. If we could see fit
8. The creative electronic musician
9. Getting Androids to learn from testimony
10. Rendering with intent
11. Deep Thought

Someday my footprints will come but we might not be quite there yet:

- The time feel might be mechanical – lack of microtiming, crucial to jazz.
- Conversation might be too polite, with its strict turn-taking.
- No personal voice – it's generalised be-bop.
- The arc is clock-driven, not music-driven.
- Motivic development might feel like repetition – because it is.
- It uses MIDI, so perhaps it inevitably sounds like a very good MIDI file and there'll be a lack of expression/articulation.
- The audience might feel it would be better musically if it used a chord progression – but the whole point is that it doesn't, it's responsive. There are plenty of play-along programs out there.